



The Propagator

The Monthly Newsletter of the South Orange Amateur Radio Association

August 2026

South Orange Amateur Radio Association | Editor: Ray Frericks, K6NOV

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Upcoming SOARA Events

This Month — August 2026

Date	Time	Event	Location
Saturday, August 15	9:00 AM–noon	SOARA Elmer Saturday	Gillieran Park
Monday, August 17	6:00 PM	Ham Radio License Exams	Murray Center
Monday, August 17	6:30 PM	Membership Meeting	Murray Center
Saturday, August 22	11:00 AM–3:00 PM	SOARA Annual Picnic	Laguna Niguel Regional Park
Monday, August 24	7:00 PM	Board Meeting	Murray Center

Next Month — September 2026

Date	Time	Event	Location
Saturday, September 19	9:00 AM–noon	SOARA Elmer Saturday	Gillieran Park
Monday, September 21	6:00 PM	Ham Radio License Exams	Murray Center
Monday, September 21	6:30 PM	Membership Meeting	Murray Center
Monday, September 28	7:00 PM	Board Meeting	Murray Center

Dates, locations, and times are subject to change. Check the [SOARA website](#) before attending.

President's Message

It is getting warm in Southern California, as you know, but it is also the perfect time to take your radio setup on the road while on vacation or maybe take a trip to a local park and do some POTA (Parks on the Air). Even though we are having some warm days, it is not as hot in Southern California as it is where I am now.

I am writing this from Defcon in Las Vegas. While Defcon is described as a "hacker convention" it is really a mixture of a lot of different interests. Interest groups are organized into "Villages." There is a Ham Radio Village, where they gave an all-day training class to get people licensed as Technician class operators. They also did license exams, with 132 new licenses and 32 upgrades. They also had fox hunts and a number of presentations on lots of ham radio topics. Defcon is very supportive of these villages and pay for exhibits, power, audio visual and the like. I believe all of the talks will eventually be available on YouTube. This year is Defcon 34, so you could search for them in a few days.

There are many other radio adjacent Villages. There is an RF Village, where people use some very nifty SDRs (and some cheap ones, like RTL-SDR) to intercept transmissions to decode them. Using GNU Radio and similar tools, they write custom demodulators to try and extract the test messages being sent. There are a lot of interesting talks on many aspects of using SDRs (software-defined radios) to decode and send using protocols like LoRa, Bluetooth and Wi-Fi. They also setup decoders for key fobs, remote control cars and other consumer devices. I am spending a lot of time learning about new ways to use GNU Radio with an SDR transceiver to try and build a LoRa Meshcore/Meshtastic router, for example.

One of the things I like about Defcon is that it allows me to sample and learn about a very wide spectrum (no pun intended) of technologies, just like amateur radio. If I want to learn more about 3D printing, there is a Village for that. If I want to learn about lock picking, there is a Village for that. Just looking at a quick list, there are at least 20 technical areas broken out that one can visit and learn from. In fact, the only problem is that with about 750 scheduled talks for the three days, it is like a firehose of information.

Amateur radio is the same way. I encourage you to get out of your comfort zone and try something new with your radio equipment. Not only will you learn something new that may expand your interest and increase your enjoyment of the hobby, it will give you something to talk about and teach to others.

SOARA is having our traditional summer picnic on Saturday, August 22 at Laguna Niguel Regional Park. We do not have an assigned area, because they are first come, first served. A couple of us will be there when the park opens on Saturday and will try to get the same area as in past years. Signs will be posted directing you, or you can call on the Temple Hill 2M repeater when you get in the park for directions.

SOARA will be grilling hamburgers, hot dogs and will have buns and condiments. We will also bring water and make lemonade. It would be nice if members who attend bring a side dish or dessert. I hope you will join us at this fun event and, of course, bring your family or significant other.

73,

Dale Tyler, W6EDT
SOARA President

SOARA Saturday Report

Thanks to everyone who came out to Gilleran Park: Ed, WA6ED; Dan, KO6BIX; Ray, K6NOV; Jim, KI6LOM; Mike, KO6AOL; Claude, W6CAG; and Greg, N6PM.

Donuts were provided, paid for by donations. We have \$2 left over for next month.

Ray, K6NOV, led a great discussion on CW, helping get folks organized so others can start practicing together. We also covered a range of other topics during the meeting.



Members gather under the shelter at Gilleran Park for July's Elmer Saturday. Photo by Ed Barnes, WA6ED.

I've been cleaning out my shack and gave away some spare parts and radio gear — expect more to give away in August.

Next meeting: Saturday, August 15, 2026, Gilleran Park, 9:00 AM–noon.

73,

Ed, WA6ED

Membership Meeting

Our August meeting is **Monday, August 17, 2026** at the **Norman P. Murray Center** in Mission Viejo. License exams start at 6:00 PM and the program begins at 6:30 PM, as usual. This month's program topic will be announced.

My Full Circle Story Adventures Thanks to Ham Radio

By Tom B. Parkes, KI6DDB

My ham radio odyssey has come full circle this 2026 Memorial Day. Ham radio introduced me to some extraordinary and very special people like our missed SOARA silent key Heiko, AD6OI. One of my earliest people helping, community service adventures was Heiko's enlisting my help for water station 6 at mile 10.3 on the Memorial Day Laguna Hills Half Marathon. After earning many Laguna Hills Half Marathon t-shirts, COVID stopped this event for a while.

Many of my pre SOARA California Triple Crown long distance cycling events were supported by ham radio nets because ham radio repeaters were the best way to cover these 200 mile events. This is why I got my Tech ticket and then joined SOARA almost 20 years ago. My venerable Yaesu VX7R and I became roving course marshals on many Cal Triple Crown cycling events.

Heiko also enlisted my ham radio support for the Orange County Marathon at the Irvine Ave. and University Dr. mile 11 medical stop. My OC Marathon staff friend, Toby Taylor, asked me to bike escort the lead male runner. My VX7R and I escorted the OC Marathon for several years.

My VX7R bike escort ham radio introduced me to emergency communications through Costa Mesa's MESAC and CERT (Community Emergency Response Team). As a full MESAC member, I'm also able to help the Huntington Beach RACES (Radio Amateur Civil Emergency Service) with their Surf City Marathon, 4th of July parade and the Pacific Air Show. Thanks to MESAC I've helped the Costa Mesa Police finish four Baker to Vegas relay race adventures. I remember SOARA members getting excited about helping Baker to Vegas. Numerous other ham radio community service and people helping opportunities opened up through MESAC.

I was "Shortcut Bike Mobile One" helping the Foothill Flyer Running club ham radio folks support the Mount Disappointment ultra marathons for several pre-COVID years.

Eventually, the SoCalBikeEscorts took over OC Marathon bike runner escorting and I became a SoCalBikeEscorts member. I'm both MESAC and SoCalBikeEscorts for the OC Marathon. Adding the SoCalBikeEscorts link to my community service chain, enabled me to support numerous non-ham radio running event adventures. Many of them are people helping charity events.

My California Triple Crown cycling - SOARA - Heiko - LH Half Marathon - OC Marathon - CM MESAC and CERT - Huntington Beach RACES - Foothill Flyer and SoCalBikeEscorts community service chain has now come full circle. This most recent 2026 Memorial Day, I once again enjoyed helping - as a non-ham radio SoCalBikeEscort - the 2026 Memorial Day Laguna Hills Half Marathon Race with a heartfelt nod to SOARA and Heiko. I have many new friends and a new Laguna Hills Half Marathon Race t-shirt to replace my mostly worn out, retired pre-COVID LHHM t-shirts.

Ham radio opens up a cornucopia of extraordinary people and community service adventures. Ham radio is a great way to thank all the people that helped all of us along the way. Ham radio is definitely for community service and people oriented adventurous spirits.

I hope SOARA members enjoy this, 73 and Cheers

Tom B. Parkes, KI6DDB

Meshcore Coaxial Sleeve Dipole (910.525 MHz): Assembly & Technical Specification

By Rich Gordon, W6BOT

Originally submitted as an 11-page technical specification dated May 3, 2026, © 2026, all rights reserved. Transcribed here preserving the author's wording; figures reproduced from the author's NanoVNA measurements and photos.

This document defines the as-built mechanical and electrical parameters for the high-stability coaxial sleeve dipole optimized for Meshcore nodes. The assembly utilizes a 7/32" brass sleeve and RG-174 coaxial cable to achieve a precise resonant match at the target frequency.

Note: AI was utilized to identify and verify dimensions, and velocity factor, for this design.

Antenna description

Theoretical basis

A sleeve dipole is a variation of the fundamental half-wave dipole. While a theoretical isotropic radiator has a gain of 0 dBi, the half-wave dipole concentrates energy perpendicular to the axis of the radiator, resulting in a gain of 2.15 dBi (some loss is expected due to velocity factor and loss tangent of the enclosure).

Factors affecting gain in this build

Common mode suppression: The 360° solder bond at the 65.5 mm mark ensures the sleeve acts as a high-impedance balun. This prevents RF current from traveling down the outside of the feedline, which would otherwise distort the radiation pattern and reduce effective gain.

Enclosure impact: The 12 mm PETG hex enclosure introduces slight dielectric losses. While PETG is relatively RF-transparent at 910.525 MHz, the proximity of the material (dielectric loading) may slightly compress the radiation pattern, potentially shifting the gain by +/-0.2–0.5 dB depending on the exact material purity and wall thickness.

Concentricity: Using the RG-58U jacket segment as a spacer ensures the radiator remains centered within the brass sleeve. Maintaining a constant characteristic impedance of 50 ohms minimizes return loss, ensuring maximum power transfer to the radiating elements.

Radiation pattern

The expected radiation pattern is toroidal (donut-shaped).

- **Maximum gain:** Perpendicular to the 150.5 mm vertical assembly (toward the horizon).
- **Null points:** Directly above the tip termination and directly below the SMA connector.

Build parameters

- **Target frequency:** 910.525 MHz.
- **Transmission line:** RG-174 coaxial cable.
- **Sleeve material:** 7/32" x 0.014" wall brass tubing.
- **Environment:** 12 mm PETG hexagonal enclosure.
- **Centering mechanism:** 2–3 mm segment of RG-58U jacket (or similar) trimmed for concentric fit.

Item	Component	Specification	Quantity
1	Coaxial Cable	RG-174 (12"/30 cm SMA jumper works well)	1
2	Sleeve Material	7/32" x 0.014" wall brass tubing — K&S Precision Metals Round Brass Tube 8130 (7/32" OD x 0.014" wall x 12" long)	60.0 mm
3	Connector	SMA crimp connector (not required if using an SMA jumper)	1
4	Centering Spacer	RG-58U jacket segment	2 pcs, 2–3 mm
5	Enclosure	PETG 3D printed (polyethylene terephthalate glycol)	1 set (2 halves)
6	Sealant	Neutral-cure silicone (e.g., Supreme Silicone M90006A)	As required

Dimension tables

All dimensions are measured starting from the 0.0 mm reference point at the top of the SMA crimp ferrule.

Individual component dimensions (as tuned)

Component	Dimension (L)	Description
Standoff Gap	5.5 mm	Distance between SMA crimp and sleeve base.
Sleeve Length	60.0 mm	Physical length of the 7/32" brass tube.
Feedpoint Gap	5.0 mm	Exposed center dielectric between sleeve and whip.
Radiator Whip	80.0 mm	Length of center conductor from top of dielectric.

Cumulative (ordinate) dimensions

Measurement Point	Ordinate (Z)	Feature Description
Reference 0	0.0 mm	Top of SMA crimp ferrule.
Point A	5.5 mm	Bottom edge of brass sleeve.
Point B	65.5 mm	Top edge of brass sleeve (360° solder point).
Point C	70.5 mm	Top of exposed center dielectric.
Point D	150.5 mm	Tip of radiating whip.



The assembled sleeve dipole, showing the 5.5 mm standoff, the 60 mm brass tube, the critical 360° solder bond, the radiator, and the tip termination. Photo by Rich Gordon, W6BOT.

Enclosure specification: 12 mm PETG hex

The antenna is housed in a custom 12 mm hexagonal PETG enclosure, providing structural protection and environmental sealing.

- **Geometry:** The hexagonal profile increases mechanical rigidity and ensures stable positioning within the node chassis.
- **Dielectric loading:** The PETG material introduces parasitic capacitance, requiring the specific tuned lengths of 60.0 mm and 80.0 mm to maintain the 910.525 MHz resonance.
- **Mechanical centering:** A 2–3 mm segment of RG-58U jacket is used as an internal spacer. This provides a precise friction fit between the RG-174 cable and the 4.85 mm ID of the 7/32" brass tube, ensuring strictly concentric geometry.

Critical assembly requirements

- **360° solder bond:** The RG-174 shield must be fully bonded to the circumference of the brass sleeve at the 65.5 mm mark to establish the RF return path. Use care to not melt the center dielectric.
- **Concentricity (zero-heuristic rule):** The RG-58U jacket spacer must be positioned to prevent the radiator from deviating from the center axis, maintaining impedance stability.
- **Dimensional sanity check:** Total dimension = 5.5 (standoff) + 60.0 (sleeve) + 5.0 (dielectric) + 80.0 (whip) = 150.5 mm.

Mechanical specification: 12 mm PETG hexagonal enclosure

The radome for the 910.525 MHz coaxial sleeve dipole is a custom-engineered hexagonal housing. It serves as both a structural chassis and a dielectric component of the antenna system.

Material requirements

- **Polymer:** PETG (polyethylene terephthalate glycol-modified).
- 3 horizontal and 3 vertical perimeters.
- **Infill:** 100% (solid wall). Air gaps within the enclosure wall will cause inconsistent dielectric loading and shift the resonant frequency.
- **Wall thickness:** Optimized for the calibrated velocity factor (Vf) of the 60.0 mm sleeve and 80.0 mm radiator.

Geometric specifications

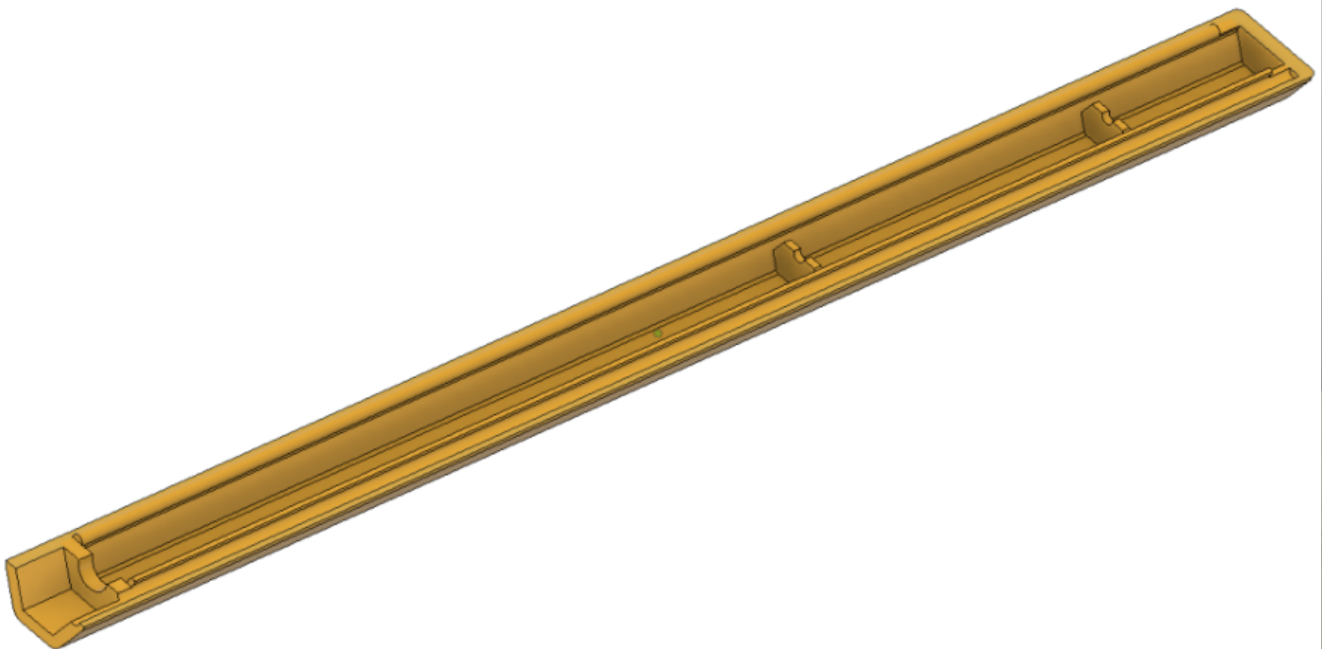
The enclosure (radome) consists of two identical halves, which snap together.

- **Profile:** Hexagonal (6-sided).
- **Width (flat-to-flat):** 12.0 mm.

- **Internal channel:** 5.80 mm to 6.00 mm diameter. This provides a clearance fit for the 7/32" (5.56 mm) brass sleeve.
- **Internal alignment:** The channel must be perfectly centered to maintain the Item #3 (concentric geometry) requirement. Off-center placement will result in an asymmetrical radiation pattern.

Dielectric considerations

- **Dielectric constant:** approximately 2.4–3.0.
- **Velocity factor shift:** The proximity of the PETG walls effectively "shortens" the wavelength. The build dimensions of 150.5 mm (total extension) are specifically tuned to compensate for this shift.
- **RF transparency:** PETG is selected for its low loss tangent at sub-GHz frequencies, ensuring minimal signal attenuation for Meshcore nodes.



3D model of the two-piece hexagonal PETG enclosure. Image by Rich Gordon, W6BOT.

Integration and sealing

- **SMA interface:** The base of the enclosure must securely capture the SMA connector body to prevent rotational stress on the 5.5 mm standoff gap.
- Should this antenna be intended for outdoor use, a moisture barrier is advisable. Once the antenna is tuned and verified, the entry and exit points of the enclosure should be sealed with a neutral-cure silicone (e.g., Supreme Silicone M90006A) to prevent internal oxidation of the brass sleeve and copper radiator.
- **Mounting:** The hexagonal flat faces allow for secure zip-tie or bracket mounting to node enclosures without the slippage common to circular radomes.
- To complete the enclosure for final assembly and make a robust connection at this high-stress point, a hex bushing designed to tightly fit over the SMA end of the enclosure. Use a heat gun or other appropriate source to warm the bushing to soften the material prior to forcing the enclosure into the bushing. Alternately, tape, wire tie or similar may be used.

Final assembly verification

The antenna must be fully inserted into the enclosure before final VNA verification.

- **Sanity check:** Ensure the 80.0 mm radiator is not bent or compressed against the end-cap of the enclosure, as this will introduce significant inductive loading.



The finished antenna mounted on a Meshcore node. Photo by Rich Gordon, W6BOT.

Tuning and calibration protocol

The following procedures define the deterministic tuning steps for verifying the resonance of the as-built antenna within the 12 mm PETG hex enclosure.

Pre-assembly resonance check

Before final sealing in the PETG housing, perform an initial VNA (Vector Network Analyzer) sweep to establish a baseline.

- **Target center frequency:** 910.525 MHz.
- **Measurement reference:** Connect the SMA interface directly to a calibrated VNA.
- **Initial SWR target:** < 1.5:1 in free air. Note that the frequency will shift downward once inserted into the PETG housing due to dielectric loading.

Physical tuning adjustments

If the resonant frequency deviates from 910.525 MHz, adjust the 80.0 mm radiator whip (Point C to Point D).

- **Frequency too low:** Trim the radiator tip in 0.5 mm increments. Apply a minimal amount of solder to the tip after each cut to prevent the RG-174 strands from unraveling.
- **Frequency too high:** Verify the 360° solder bond at the top of the sleeve (Point B). A cold joint or incomplete bond at the 65.5 mm mark will cause erratic resonance shifts.
- **Impedance matching:** Ensure the RG-58U jacket spacer is maintaining concentricity between the RG-174 and the 7/32" brass tube. Non-concentric geometry shifts the characteristic impedance away from 50 ohms.

As you can see by the graphs, while the 2:1 bandwidth is fairly wide, there is a significant advantage in tuning to your desired frequency. Note however that the wavelength at this frequency is approximately 330 mm / 12.96". When tuning, it is critical that you make your trim based on readings with the enclosure installed.

Threshold	Low Frequency (fL)	High Frequency (fH)	Total Bandwidth (Δf)
2.0:1	~ 887.0 MHz	> 935.0 MHz	> 48.0 MHz
1.5:1	~ 898.5 MHz	~ 922.5 MHz	24.0 MHz

Enclosure calibration

The 12 mm PETG hex enclosure acts as a radome and alters the electrical length of the antenna.

- **Dielectric offset:** Final tuning must be performed with the antenna fully seated in the enclosure.
- **Final verification:** The as-built total extension of 150.5 mm (from SMA crimp to radiator tip) is specifically calibrated for the velocity factor shift introduced by this housing.

Summary of tapping/ordinate points for VNA reference (910.525 MHz)

Measurement Goal	Component to Inspect	Ordinate (Z) Target
Coarse Tuning	Radiator Whip Length	80.0 mm
Return Loss/SWR	360° Solder Joint	65.5 mm
Stability/Noise	Standoff Gap	5.5 mm
Resonance Center	Total Extension	150.5 mm

Alternate Meshcore frequencies utilized in the US

Frequency (MHz)	Standoff Gap	Sleeve Length	Feedpoint Gap	Radiator Whip	Total Length
909.875 (Sacramento)	5.50	60.04	5.00	80.06	150.60
910.525 (Reference)	5.50	60.00	5.00	80.00	150.50
915.000 (High Speed)	5.47	59.71	4.98	79.61	149.77
927.875 (SoCal)	5.40	58.88	4.91	78.50	147.69

As you can see from the table, the minor adjustments for these frequencies are beyond what would be practical for "cut to length," and due to variations in materials each antenna needs to be fine tuned with a VNA for the actual frequency.

Dimensions for other frequencies may be calculated by: $L_{NEW} = L_{REF} \times (f_{REF} / f_{NEW})$.

Other notes and comments

My decision to build this antenna stems from the poor performance of the 40 mm stub antennas which come packaged with the Meshcore "Starter" kits, and the fake claims of high gain whips typically available from Amazon and AliExpress.

Keep in mind that a 1/2 wave dipole has a gain of 2.15 dBi (gain over an isotropic radiator) due to the radiation being shaped into a "donut."

Antenna Type	Typical Marketed Gain	Actual Realized Gain	Inflation Delta (Δ)	Technical Cause of Delta
40 mm Kit Stub	2.5 dBi	-1.5 dBi	4.0 dB	High ohmic loss in loading coil; aperture too small.
82 mm 1/4 λ Whip	3.0 dBi	0.5 – 1.0 dBi	2.0 dB	Missing ground plane / counterpoise in field use.
1/2 λ Sleeve Dipole (this build)	1.9 dBi	1.9 dBi	0.0 dB	Verified via NanoVNA; ground-independent.
190 mm Whip	10 dBi	1.5 – 2.5 dBi	4.85 – 7.85 dB	Mathematically impossible.
18" Collinear	8.0 – 10.0 dBi	5.5 – 6.5 dBi	2.5 – 3.5 dB	Phasing coil losses; pattern "lobing" (energy tilt).
600 mm Yagi	18.0 dBi	9.0 – 11.0 dBi	7.0 – 9.0 dB	Physical size restricts element count (aperture limit).

The "cable loss" oversight

A common fallacy across all antenna types is ignoring the feed line. RG-58 or RG-174 coax has high attenuation at 910 MHz.

- **RG-174 loss:** Approximately 1.0 dB per meter (0.3 dB/ft).
- **The result:** If you use a "high-gain" 6 dBi antenna but connect it with 3.0 meters (9.8 ft) of RG-174, you have lost 3.0 dB in the cable. Your net system gain is now 3.0 dBi, roughly the same as mounting this sleeve dipole directly to the node with no cable.

If you would like to build this antenna I can provide the STL files for the enclosure. Please send any comments, questions or requests for the enclosure STL files to wsixbot@gmail.com.

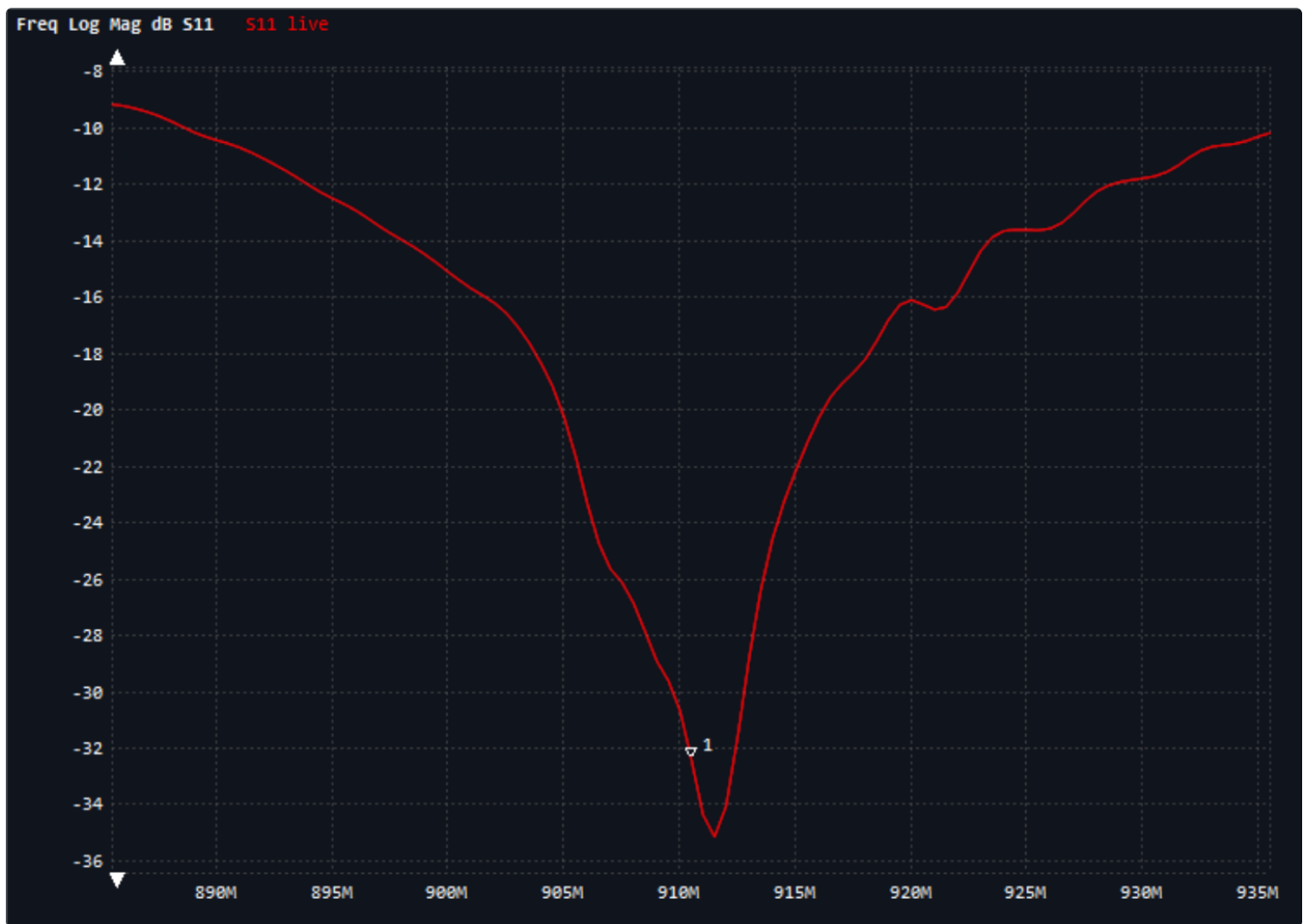
73, Rich Gordon, W6BOT

Measurement plots (NanoVNA)

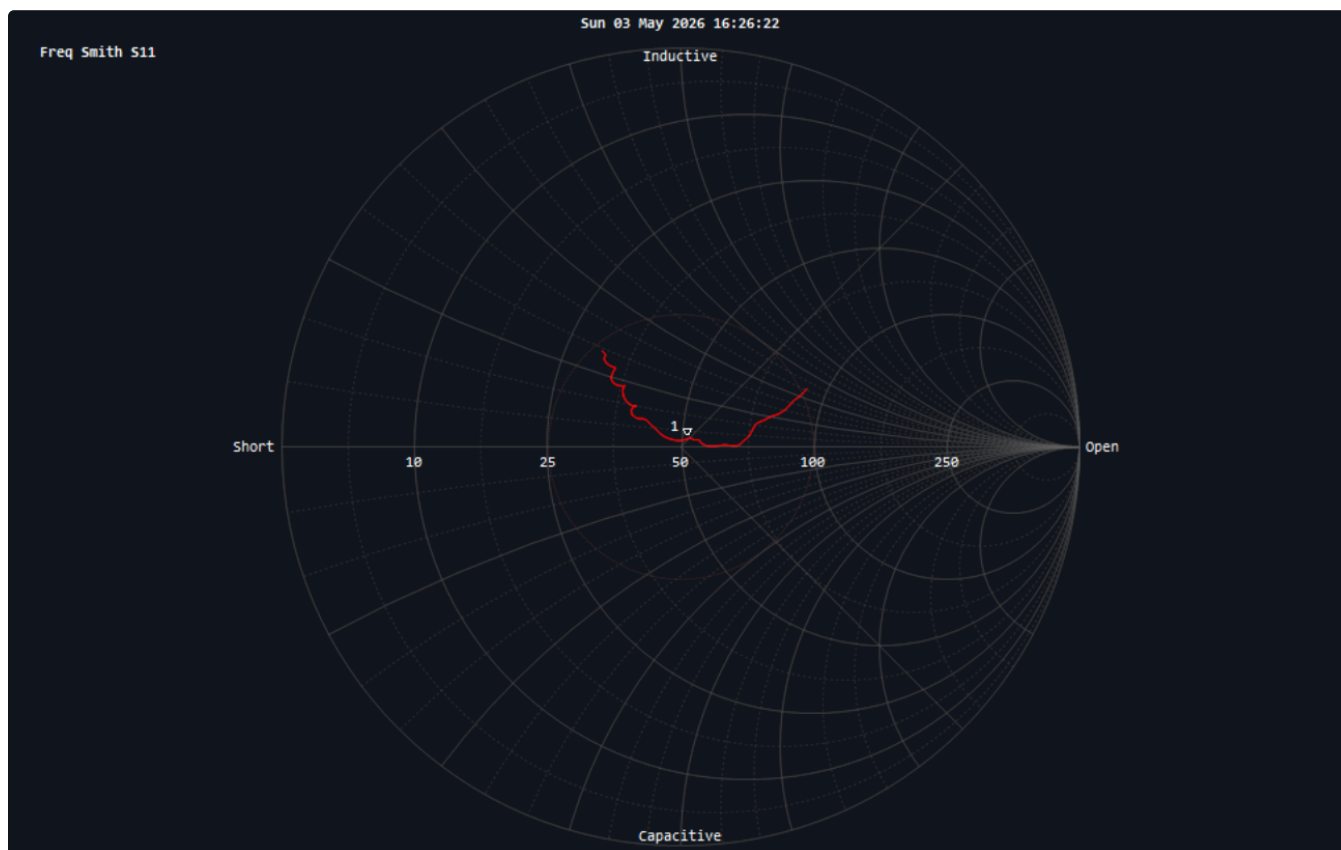
The author supplied the following NanoVNA measurements of the as-built antenna.



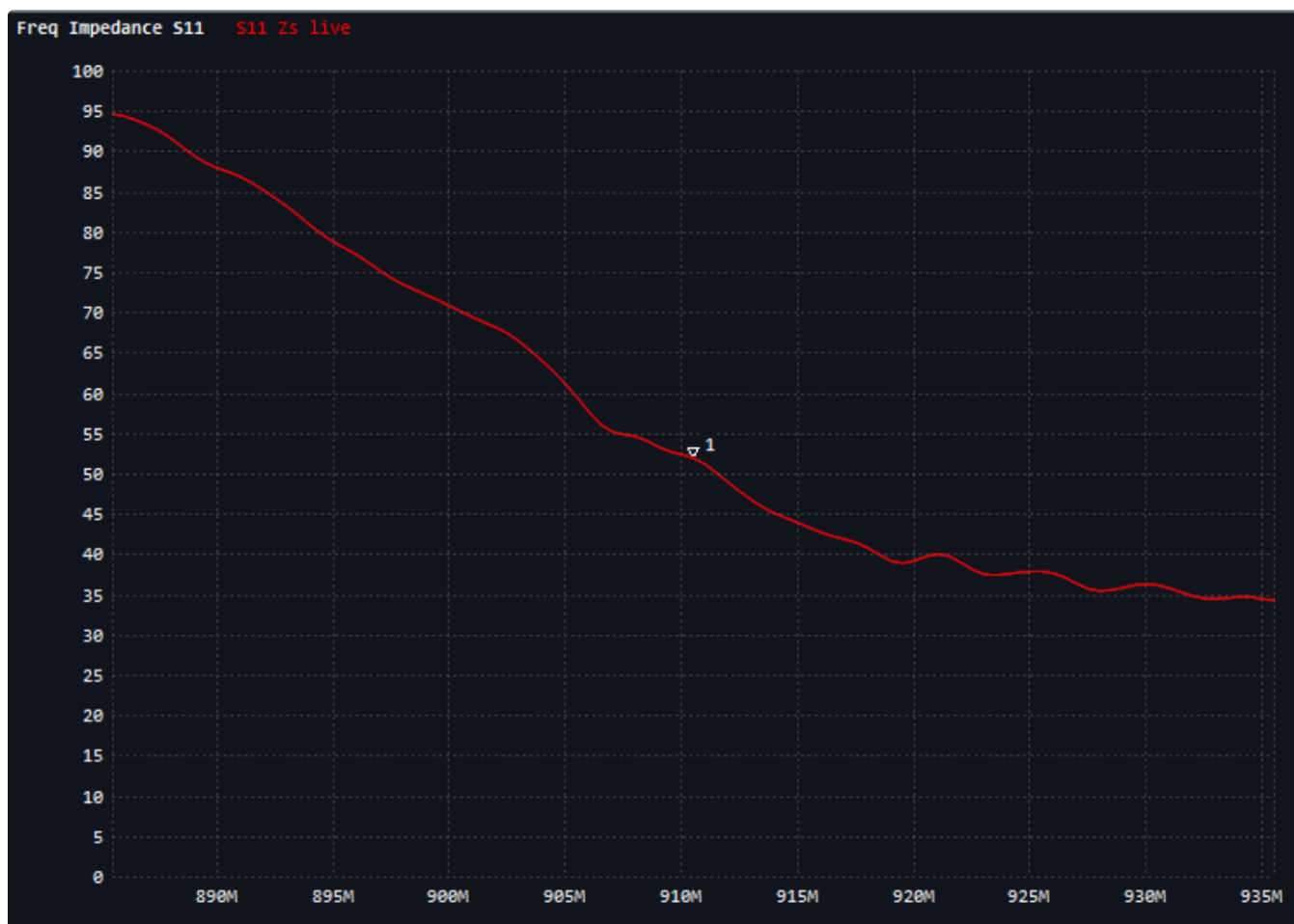
Measured VSWR sweep. The SWR minimum sits near the target frequency with a wide 2:1 bandwidth. NanoVNA plot by Rich Gordon, W6BOT.



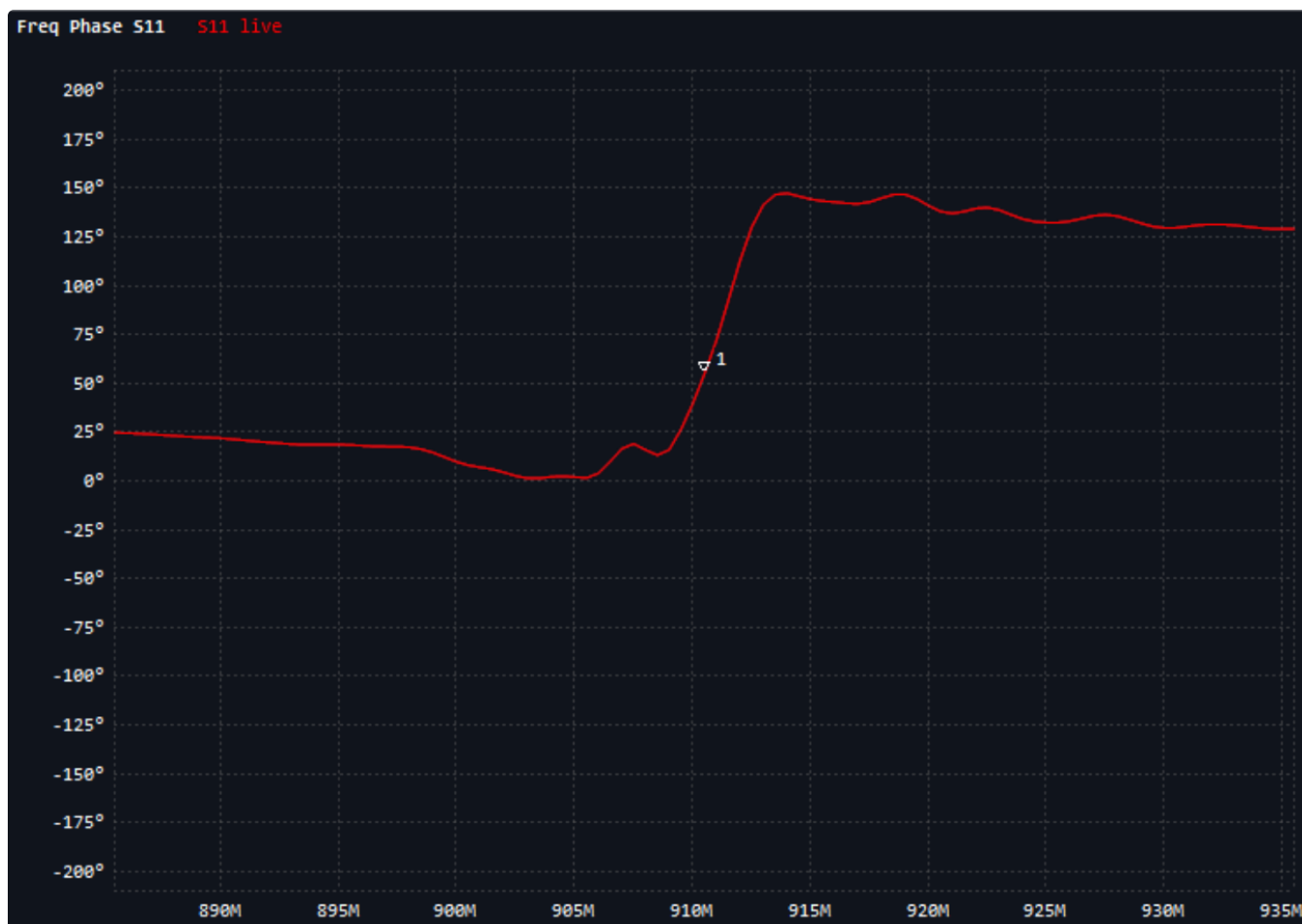
S11 return loss, log magnitude. NanoVNA plot by Rich Gordon, W6BOT.



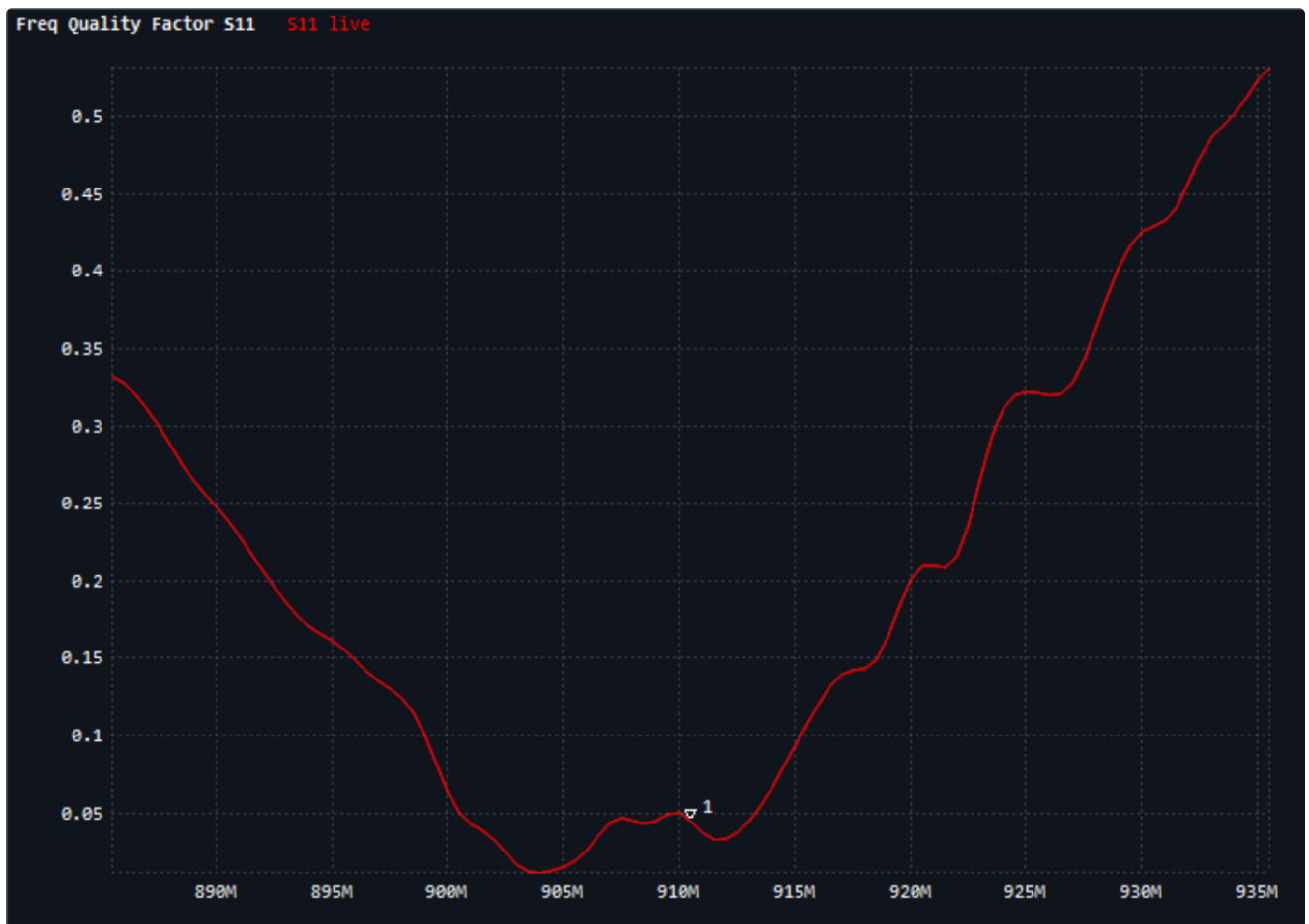
Smith chart showing the impedance locus through resonance. NanoVNA plot by Rich Gordon, W6BOT.



Impedance versus frequency. NanoVNA plot by Rich Gordon, W6BOT.



Phase versus frequency. NanoVNA plot by Rich Gordon, W6BOT.



Quality factor (Q) across the sweep. NanoVNA plot by Rich Gordon, W6BOT.

Treasurer's Report

Reporting period: October 1, 2025 through July 31, 2026 (ten months)

Prepared by: Ron Mosher, K0PGE, SOARA Treasurer

Ten Months	July 2026	July 2025	Increase/(Decrease)
Cash Income:			
Memberships	\$8,315.00	\$9,781.00	\$(1,466.00)
Other & Interest	\$2,370.49	\$3,848.00	\$(1,477.51)
Total Cash Income	\$10,685.49	\$13,629.00	\$(2,943.51)
Cash Expenses:			
Repeaters incl. site rental, utilities & insurance	\$6,158.83	\$6,141.00	\$17.83
Member Activities	\$2,995.71	\$2,966.00	\$29.71
Other expenses	\$2,822.58	\$1,217.00	\$1,605.58
Total Cash Expenses	\$11,977.12	\$10,324.00	\$1,653.12
Cash Net Income	\$(1,291.63)	\$3,305.00	\$(4,596.63)
Beginning Cash — October 1	\$28,179.03	\$25,745.00	\$2,434.03
Ending Cash — July 31	\$26,887.40	\$29,050.00	\$(2,162.60)

If any member has questions about the financials, please contact the Treasurer or any Board member.

SOARA Information

Meetings: SOARA meets at the Norman P. Murray Center, 24932 Veterans Way, Mission Viejo, CA on the third Monday of every month at 7:00 PM. For January and February, the third Monday is a holiday and the meeting is held on the fourth Monday.

License Exams: Amateur license exams are given prior to SOARA meetings, except June. Exams are at 6:00 PM. Prior registration is not required and walk-in applicants are welcome. For June, exams are held at Field Day. For further information, email Steve Kuver, K6UVR, at k6uvr@soara.org.

SOARA Library: SOARA has many amateur radio related books such as handbooks, books about electrical theory, and more available to lend to club members. Contact library@soara.org for more information.

Web Site: SOARA maintains a web site with current club information at soara.org.

Repeaters

The Laguna Beach, San Clemente, and Trabuco repeaters are open. The Santiago Peak repeaters are closed. For details or questions on the repeaters, contact the repeater director at repeater@soara.org.

Band	Frequency	Shift	Tone / Notes	Location
2m	147.645	-	(110.9)	Laguna Beach
2m	146.025	+	(110.9)	San Clemente
2m	145.240	-	(110.9)	Trabuco
D-STAR 2m	146.115	+	(K6SOA C)	Laguna Beach
220	224.100	-	(110.9)	Laguna Beach
220	224.640	-	(pvt)	Santiago Peak (C)
440	445.660	-	(110.9)	Laguna Beach
D-STAR 440	445.705	-	(K6SOA B)	Laguna Beach
440	447.180	-	(pvt)	Santiago Peak (C)
D-STAR 1.2G	1282.600	-	(K6SOA A)	Laguna Beach

Nets

- 40 meter HF (7.200 MHz +/-): Sundays at 8:00 AM
- 10 meter HF (Technicians welcome) (28.415 +/-): Sundays at 9:00 AM
- General Membership Net — UHF/VHF (447.180, 147.645 & 224.640): Tuesdays at 8:00 PM

- D-STAR (146.115 C module): Wednesdays at 8:00 PM
- Tech Net — 147.645, 224.640, 447.180: Saturdays at 9:00 AM
- California Rescue Communications (Gordo Net) HF (7.250 MHz +/- for QRM): Weekdays at 8:30 AM
- MVRACES — 447.180: Tuesdays at 7:00 PM
- Tri-Cities RACES — 146.025: Wednesdays at 8:00 PM
- LNACS — 147.645: Thursdays at 7:00 PM
- OC Parks Fire Watch — 447.180: Thursdays at 8:00 PM

SOARA Officers

- **President:** Dale Tyler, W6EDT — w6edt@soara.org — 949-360-1717
- **Vice President:** Greg Unruh, WE4BY — we4by@soara.org
- **Secretary:** Charles Schultz, NY6I — ny6i@soara.org
- **Treasurer:** Ron Mosher, K0PGE — k0pge@soara.org — 949-363-0047

SOARA Directors

- **Repeater:** Open
- **Publications:** Ray Frericks, K6NOV — k6nov@soara.org
- **Membership:** Open
- **Education:** Steve Kuver, K6UVR — k6uvr@soara.org — 949-874-1972
- **Technical:** Tom Hobbs, AE6SH — ae6sh@soara.org — 949-887-6527
- **Communications:** Ray Hutchinson, AE6H — ae6h@soara.org

SOARA Appointments

- **Past President:** Ray Hutchinson, AE6H — ae6h@soara.org
- **Activities Coordinator:** Open
- **SOARA Saturdays:** Ed Barnes, WA6ED — wa6ed@soara.org — 949-468-7706
- **Raffle:** Open
- **Testing:** Steve Kuver, K6UVR — k6uvr@soara.org
- **Website:** Open

Contacting SOARA

Website: soara.org | Membership: membership@soara.org | Postal: P.O. Box 2545, Mission Viejo, CA 92690 | Facebook: K6SOA | Twitter/X: @K6SOA

Dates, locations, times, and frequencies are subject to change. Check the SOARA website before attending an event.

Questions about SOARA? membership@soara.org | P.O. Box 2545, Mission Viejo, CA 92690 | soara.org | Facebook/X: K6SOA